

Adith Abhilash

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Professional Summary

Software and AI engineer building production systems across backend infrastructure, LLM applications, and computer vision. Recent work spans a geospatial scoring platform for EV infrastructure planning, a citation-grounded RAG pipeline, and CNN-based perception for autonomous edge devices. Comfortable moving between backend architecture and model development on the same project, with a strong foundation in systems design, data structures, and applied machine learning.

Technical Skills

- **Languages & Systems:** Python, C++, C, SQL, React JS, JavaScript, HTML/CSS
- **Backend & Infrastructure:** FastAPI, Flask, RESTful API design, PostgreSQL, MySQL, Git
- **AI/ML:** LLM application development (RAG, prompt engineering, context management), computer vision (CNNs, OpenCV), PyTorch, TensorFlow, NLP
- **Applied Domains:** Geospatial systems, OCR pipelines, edge computing (Jetson Nano, Raspberry Pi, Pixhawk 6X)

Experience

Software Developer Intern, Hound Mobility

Jun 2026 – Present

- Engineered a geospatial web application (Python, FastAPI, Google Maps API) to collect, process, and visualize spatial data for prospective EV charging station locations across more than 100 sites per region.
- Designed a weighted scoring model evaluating about 10 site parameters to generate feasibility scores for EV charging site selection, reducing manual evaluation time by 60% used to shortlist 100 sites.
- Built scalable RESTful backend APIs for real-time site data extraction, processing, and score visualization, handling more than 100 requests/records.

Software & AI Intern, AGH Advisors

Jun 2025

- Developed an automated news aggregation pipeline using Google News feeds and NLP-based sentiment analysis, classifying 1000's articles by sentiment with 80% accuracy.
- Implemented an OCR document-processing pipeline to extract text from more than 20 scanned PDF question papers, achieving 70-80% extraction accuracy and reducing manual transcription effort.
- Applied NLP classification to categorize extracted questions by Bloom's Taxonomy level, achieving 90% classification accuracy.

Projects

LLM Token-Based Memory Management | Python, LLMs, NLP

- Designed a token-aware LLM context management system that dynamically prunes conversation history, reducing token consumption by 50% while preserving response quality across 1000's of turn conversations.

RAG System with Citation | Python, RAG, Vector Databases

- Architected a Retrieval-Augmented Generation pipeline over 50+ documents using vector DB/framework (FAISS, LangChain) for document-grounded question answering with inline source citations, improving answer traceability and reducing unsupported responses.

Autonomous Drone System — Hoverlords, SJ CET | Computer Vision, CNNs, Jetson Nano, Raspberry Pi, Pixhawk 6X

- Built CNN-based perception and statistical control systems for autonomous drone delivery and agricultural spraying on edge hardware, achieving 90% detection accuracy across more than 10 test flights.

Pandemic Spread Simulator | Neural Networks, GAMA Platform

- Trained a neural network to predict virus propagation dynamics and integrated predictions with an agent-based spatial simulation modeling 5 agents in GAMA platform.

Library Management System | Python, SQL, MySQL

- Built a database-driven application implementing CRUD operations, transaction tracking, and automated record logging with relational database design across 5 tables.

Education

B.Tech in Artificial Intelligence and Data Science – St. Joseph's College of Engineering and Technology, Palai – CGPA: 7.02/10

August 2022 – May 2026

Higher Secondary (12th Grade) – Chavara CMI Public School, Palai – Percentage: 74%

June 2020 – March 2022

Secondary (10th Grade) – St. Vincent CMI Residential School, Palai – Percentage: 81%

– March 2020